

Take-Home Assignment:

Incident Investigation Agentic AI Application

Duration: 7 Days

Objective

Build a working Agentic AI Assistant that helps equipment engineers investigate machine downtime incidents. The assignment focuses on practical implementation, engineering quality, and effective use of LLMs.

Problem Statement

A semiconductor manufacturing plant wants to build an AI assistant to support equipment engineers during machine downtime investigations. Engineers currently gather information manually from multiple systems including equipment records, alarm history, maintenance records, previous incidents and SOPs before determining the next course of action.

Your task is to build an Agentic AI Assistant that retrieves relevant information from the provided dataset, reasons over the available context using a real LLM, and generates a draft investigation report.

Assignment Scope

- Accept a free-text incident description.
- Retrieve relevant information from the provided dataset.
- Use a real LLM to analyse the incident.
- Generate recommended actions and escalation advice.
- Produce a structured investigation report.

Minimum Functional Requirements

- Simple Web UI, CLI or REST API.
- Tool calling to access the provided dataset.
- Real LLM API integration.
- Structured incident report generation.
- Clear and modular code structure.

Design Document

- Overall solution architecture.
- Agent workflow and orchestration.

- Tool design.
- Prompting strategy.
- Key implementation decisions and assumptions.

Testing

Provide at least four test scenarios:

- Normal incident
- Missing information
- Repeated incident
- Unknown alarm

Submission Requirements

Please submit the following:

- Source code hosted in a GitHub repository.
- Share the GitHub repository link with us (public repository or private repository with reviewer access).
- Design document (PDF or Word).
- README containing setup instructions and how to run the application.
- Sample prompts and test cases.
- Short demo video demonstrating the solution.

Optional Enhancements

- Multi-agent workflow
- Conversation memory
- Vector search
- Reflection/self-review
- Streaming responses
- Docker deployment
- Evaluation metrics

What We Are Looking For

- Code quality and maintainability
- Agentic AI workflow design
- Tool integration
- Effective LLM usage
- Software engineering practices

- Clear documentation
- Thoughtful design trade-offs